

ON THE DIFFRACTION SPECTRUM OF THE SET OF VISIBLE POINTS IN LATTICES AND CERTAIN CUT-AND-PROJECT SETS

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ABSTRACT. Let $k \geq 2$ be a positive integer. It is known that the set of visible lattice points from the origin in $\mathbb{Z}^k$ has a translation bounded pure point diffraction spectrum. We investigate these properties for the set of visible points from a *finite set of locations* $\{x_1, \dots, x_n\} \subseteq \mathbb{Z}^k$. We provide explicit formulas for the coefficients of the diffraction spectrum. Additionally, we generalize our procedure to show that the set of visible points in certain classes of cut-and-project sets has a translation bounded diffraction pure point spectrum.

1. INTRODUCTION

Let $\Lambda \subset G$ be a uniformly discrete subset of a locally compact σ -compact Abelian group. One approach to studying the structure of Λ is through the analysis of the *autocorrelation* and its *diffraction spectrum* associated with the *Dirac comb* measure

$$\omega_\Lambda \stackrel{\text{def}}{=} \sum_{\mathbf{x} \in \Lambda} \delta_{\mathbf{x}},$$

which is a translation bounded measure (see [BL04, Lemma 2] for a general locally compact σ -compact Abelian group G and [BG13, Example 8.6] in the Euclidean case). To define the autocorrelation and the diffraction spectrum, one must make several choices, including an increasing sequence $\mathcal{A} = \{A_n\}_{n \in \mathbb{N}}$ of measurable subsets of G and a normalized Haar measure θ_G . We denote by $\omega_\Lambda|_{A_n}$ the restriction of ω_Λ to the set A_n and denote by $\widetilde{\omega}_\Lambda$ the twisted version of ω_Λ ; see eq. (9) for the definition. An *autocorrelation* is a weak-* accumulation point of the family of convolutions

$$\left\{ \frac{\omega_\Lambda|_{A_n} * \widetilde{\omega_\Lambda|_{A_n}}}{\theta_G(A_n)} : n \in \mathbb{N} \right\}$$

as $n \rightarrow \infty$. When this family has a unique accumulation point, that measure is called the *autocorrelation* of ω_Λ and is denoted by γ_{ω_Λ} . The precise choices of these objects will be specified later in the context of cut-and-project sets. The *diffraction spectrum* of Λ is the Fourier transform of the autocorrelation,

$$\widehat{\gamma_{\omega_\Lambda}}.$$

If the support of the diffraction spectrum is discrete, we say that the diffraction spectrum is *pure point*. For instance, appropriate choices allow one to recover the density of the set Λ , namely,

$$\widehat{\gamma_{\omega_\Lambda}}(0) = \theta(\Lambda).$$

Provided that the limit exists, the *density* of Λ is defined as the limit of the proportion of elements in $\Lambda \cap A_n$ relative to the Haar measure of A_n ; see (5) for the formal definition. For more details on diffraction spectra, see [HR15, Sch00, BMP00, Hof95] and the references therein.

1.1. Results. We now describe the main results of this paper. We begin with the classical case of visible points in the integer lattice $\mathbb{Z}^k$, which motivates our study of points that are simultaneously visible from a finite set of lattice points and, later, visible points in a class of cut-and-project sets.

1.1.1. Visible points in the integer lattice. Let $k \geq 2$ be a positive integer. Denote the Dirac comb of $\mathbb{Z}^k$ by

$$\omega_{\mathbb{Z}^k} = \sum_{\mathbf{x} \in \mathbb{Z}^k} \delta_{\mathbf{x}}.$$

By the Poisson summation formula, the diffraction spectrum is pure point and is given by

$$\widehat{\gamma_{\omega_{\mathbb{Z}^k}}} = \omega_{\mathbb{Z}^k}.$$

In particular, the coefficients of the diffraction spectrum are constant and equal to 1. Physically, if X-rays (photons of a fixed wavelength) are transmitted through $\mathbb{Z}^k$, one observes a periodic array of bright spots known as *Bragg peaks*. These Bragg peaks are supported on $\mathbb{Z}^k$, and each peak has unit intensity.

Given two distinct points $\mathbf{x} = (x_1, \dots, x_k)$ and $\mathbf{y} = (y_1, \dots, y_k)$ in $\mathbb{Z}^k$, we say that $\mathbf{y}$ is *visible* from $\mathbf{x}$ if no other point of $\mathbb{Z}^k$ lies on the line segment joining them. Let V_k denote the set of points in $\mathbb{Z}^k$ that are visible from the origin, namely,

$$V_k \stackrel{\text{def}}{=} \{(x_1, \dots, x_k) \in \mathbb{Z}^k : \gcd(x_1, \dots, x_k) = 1\}.$$

By the Chinese Remainder Theorem, the set $\mathbb{Z}^k$ contains arbitrarily large visible-point-free cubes (cf. [Apo76, Theorem 5.29] for the integer lattice, and [BMP00, Proposition 4] for arbitrary full-rank lattices). Thus, despite being uniformly discrete, V_k fails to be relatively dense and hence is not a Delone set (see Section 2.3). In fact, since these holes repeat periodically throughout the lattice, adjoining any set of zero density to V_k still does not produce a Delone set.

For $k \geq 2$, Baake–Moody–Pleasant [BMP00, Theorem 3] studied the diffraction spectrum $\widehat{\gamma_{\omega_{V_k}}}$ of the set $V_k \subset \mathbb{Z}^k$. They showed that $\widehat{\gamma_{\omega_{V_k}}}$ is again a translation bounded pure point measure. Moreover, the diffraction spectrum can be computed explicitly and is supported on the subset of $\mathbb{Q}^k$ consisting of vectors with square-free denominators and is given by

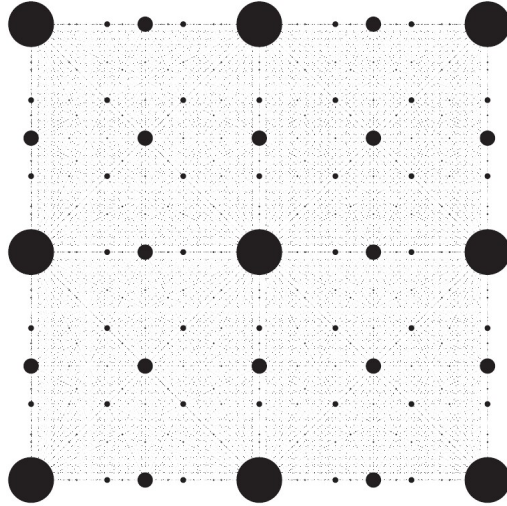
$$\widehat{\gamma_{\omega_{V_k}}} = \sum_{\substack{\mathbf{u} = \frac{\mathbf{m}}{N} \in \mathbb{Q}^k \\ \mathbf{m} \in \mathbb{Z}^k \\ N \text{ square-free}}} |a(\mathbf{u})|^2 \delta_{\mathbf{u}}, \quad a(\mathbf{u}) = \frac{1}{\zeta(k)} \prod_{p|N} \frac{1}{1 - p^k}.$$

In fact, they established this result in the more general setting of arbitrary full-rank lattices in dimension $k \geq 2$, as well as for the one-dimensional set of k th-power-free integers. See Figure 1 for an illustration of the diffraction spectrum of visible lattice points in $\mathbb{Z}^2$. The study of diffraction spectra for visible lattice points and k -free lattice points has been further extended to $\mathcal{B}$ -free lattice systems (see, for example [BLS26, § 4], [BHS17, Corollary 21] and reference therein). This motivates the following question. Let $S \subset \mathbb{Z}^k$ be a finite set.

Question. What can be said about the diffraction spectrum of the points in $\mathbb{Z}^k$ that are simultaneously visible from every element of S ?

Throughout the present work, we write $\#(A)$ for the cardinality of a set A . For each $p \in \mathcal{P}$, the map π_p denotes the natural projection

$$(1) \quad \pi_p : \mathbb{Z}^k \longrightarrow (\mathbb{Z}/p\mathbb{Z})^k, \quad (x_1, \dots, x_k) \longmapsto (x_1 \bmod p, \dots, x_k \bmod p),$$


 FIGURE 1. Diffraction of the visible points of $\mathbb{Z}^2$ [BG13, p. 424].

and

$$s(p) \stackrel{\text{def}}{=} \#(\pi_p(S))$$

is the cardinality of the image of S under π_p . Define $V(S) \subseteq \mathbb{Z}^k$ to be the set of integer lattice points that are simultaneously visible from all points of S . Rumsey [Rum66, Theorem 7] proved that the density of the points simultaneously visible from S , averaged over the boxes $[-n, n]^k$, is given by

$$(2) \quad \theta(V(S)) = \prod_{p \in \mathcal{P}} \left(1 - \frac{s(p)}{p^k}\right),$$

where $\mathcal{P}$ denotes the set of prime numbers. Rumsey extended this formula to certain infinite subsets S satisfying appropriate conditions and observed that the same density formula holds when averaging over rectangular boxes

$$[-n_1, n_1] \times \cdots \times [-n_k, n_k],$$

as $\min\{n_1, \dots, n_k\} \rightarrow \infty$. We refer the reader to [BKP23] for more details on simultaneous visibility and references therein.

In the present work, we consider the Dirac comb measure

$$\omega_{V(S)} \stackrel{\text{def}}{=} \sum_{\mathbf{x} \in V(S)} \delta_{\mathbf{x}}$$

and prove the following.

Theorem 1.1. *Let $S \subset \mathbb{Z}^k$ be a finite set with $k \geq 2$ and assume that the density of $V(S)$, computed along the averaging sets $[-L, L]^k$, is strictly positive. Then*

- (1) *The diffraction spectrum $\widehat{\gamma_{\omega_{V(S)}}$ of the set of visible points $V(S)$ is a pure point translation-bounded measure;*
- (2) *More precisely,*

$$\widehat{\gamma_{\omega_{V(S)}}} = \sum_{\substack{\mathbf{u} = \frac{\mathbf{m}}{N} \in \mathbb{Q}^k \\ \mathbf{m} \in \mathbb{Z}^k \\ N \in \mathbb{N} \text{ square-free}}} |a(\mathbf{u})|^2 \delta_{\mathbf{u}},$$

where

$$a(\mathbf{u}) = \theta(V(S)) \prod_{\substack{p \in \mathcal{P} \\ p|N}} \frac{\left(-\frac{1}{p^k} \sum_{\mathbf{x} \in \pi_p(S)} \exp \left(-2\pi i \frac{\langle \mathbf{v}_p, \mathbf{x} \rangle}{p} \right) \right)}{\left(1 - \frac{s(p)}{p^k} \right)}, \quad \mathbf{v}_p = -\mathbf{m} \pmod{p}.$$

We explain the notation and prove these results in Sections 3.1 and 3.2.

1.1.2. *Visible points in cut-and-project sets.* Let d, m be positive integers. Let $\mathcal{L} \subset \mathbb{R}^d \times \mathbb{R}^m$ be a full-rank lattice. Given a measurable set $\mathcal{W} \subseteq \mathbb{R}^m$, called a *window*, the cut-and-project set $\Lambda(\mathcal{L}, \mathcal{W})$ is the projection onto $\mathbb{R}^d$ of the lattice points that belong to the strip $\mathbb{R}^d \times \mathcal{W}$, that is,

$$\Lambda(\mathcal{L}, \mathcal{W}) \stackrel{\text{def}}{=} \{\mathbf{x} \in \mathbb{R}^d : (\mathbf{x}, \mathbf{y}) \in \mathcal{L}, \mathbf{y} \in \mathcal{W}\}.$$

We postpone the technical details to the following sections and first place our results in the context of the existing literature.

The Dirac comb associated with the cut-and-project set $\Lambda \stackrel{\text{def}}{=} \Lambda(\mathcal{W}, \mathcal{L})$ is defined by

$$\omega_\Lambda \stackrel{\text{def}}{=} \sum_{\mathbf{x} \in \Lambda} \delta_{\mathbf{x}}.$$

The diffraction spectrum $\widehat{\gamma_{\omega_\Lambda}}$ of an irreducible cut-and-project set is a translation bounded pure point measure, provided that $\mathcal{W}$ is compact. See [BG13, Theorem 9.4] and [Hof95] for the Euclidean case. For the general setting of locally compact Abelian groups, see [Sch00, BM04]. Moreover, it admits an explicit description in terms of the Fourier transform of the window. See Figure 2 for the Ammann–Beenker point set and its diffraction image.

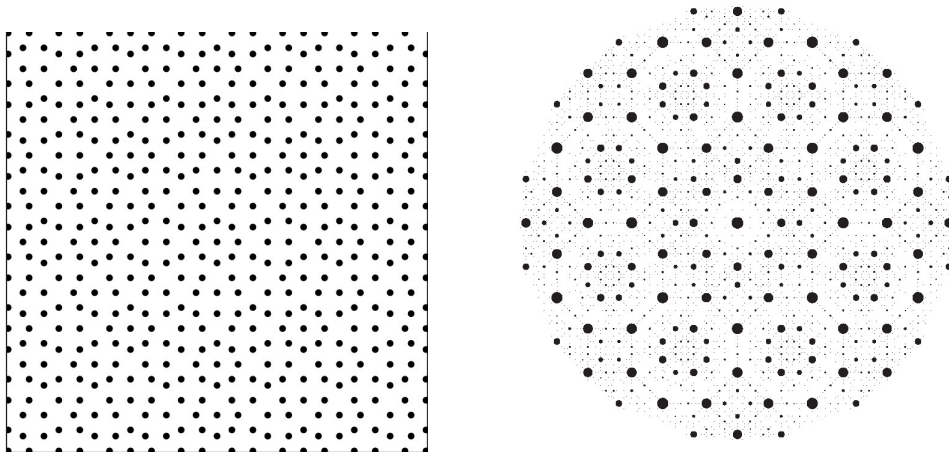


FIGURE 2. Ammann–Beenker point set (left) and its diffraction image (right), adapted from [Sin] and [BG13, p. 372], respectively.

Denote by $\Lambda_{\text{vis}} \subset \Lambda(\mathcal{L}, \mathcal{W}) \setminus \{\mathbf{0}\}$ the set of points that are visible from the origin. Marklof and Strömbergsson [MS15, Theorem 1] proved that for any irreducible cut-and-project set, the density of Λ_{vis} exists and satisfies

$$(3) \quad 0 < \theta(\Lambda_{\text{vis}}) \leq \theta(\Lambda).$$

On the other hand, Hammarhjelm [Ham22] constructed examples of cut-and-project sets $\Lambda(\mathcal{L}, \mathcal{W}) \subseteq \mathbb{R}^2$ for which the inequality in (3) is strict and computed $\theta(\Lambda_{\text{vis}})$ explicitly. Some of these examples were also considered by Sing [Sin]. More recently, in [GKW25], the first-named author together with I. Gringlaz and B. Weiss extended Hammarhjelm's

density results to all dimensions $d \geq 2$ and established an error term. Moreover, [GKW25, Theorem 1.1] implies that Λ_{vis} contains arbitrarily large holes.

Theorem 1.2. [GKW25, Theorem 5.1] *Let $d \geq 2$, let $m = d$ and let $\mathcal{W} \subset \mathbb{R}^d$ be a convex centrally symmetric window. Let K be a real quadratic field satisfying the Hammarhjelm condition and let*

$$(4) \quad \mathcal{L} \stackrel{\text{def}}{=} \{(x_1, \dots, x_d, \sigma(x_1), \dots, \sigma(x_d)) : x_i \in \mathcal{O}_K\} \subset \mathbb{R}^{2d},$$

where σ is the non-trivial Galois automorphism of K .

Let $\Lambda = \Lambda(\mathcal{W}, \mathcal{L})$ and let $D \subset \mathbb{R}^d$ be a convex averaging set. Then, as $T \rightarrow \infty$,

$$\frac{\#(\Lambda_{\text{vis}} \cap TD)}{\text{vol}(TD)} = \left(1 - \frac{1}{\lambda^d}\right) \cdot \frac{\theta(\Lambda)}{\zeta_{\mathcal{O}_K}(d)} + \begin{cases} O\left(\frac{\log T}{\sqrt{T}}\right), & d = 2, \\ O\left(\frac{1}{\sqrt{T}}\right), & d \geq 3. \end{cases}$$

We explain the notation of the previous result in Section 2.1. In this paper, we consider cut-and-project sets arising from algebraic lattices over quadratic number fields satisfying the Hammarhjelm condition; see Section 2.1 for the definition. We prove the following.

Theorem 1.3. *Let K be a real quadratic field satisfying the Hammarhjelm condition. Let $\mathcal{W} \subset \mathbb{R}^d$ be a convex centrally symmetric window, and let $\mathcal{L} \subset \mathbb{R}^{2d}$ be the lattice defined in (4). Let $\Lambda = \Lambda(\mathcal{W}, \mathcal{L})$ be the associated cut-and-project set and let $D \subset \mathbb{R}^d$ be a convex Jordan measurable averaging set. Then the diffraction spectrum $\widehat{\gamma_{\Lambda_{\text{vis}}}}$ is a translation bounded pure point measure along the sequence $\{TD\}_{T>0}$.*

We prove the statement in Section 3.3. The main ingredients are the density estimates from [GKW25] and Baake, Huck, and Strungaru's results on the diffraction of weak model sets.

1.2. Ideas of the proofs. Our methods rely on establishing a connection between sets of visible points and weak model sets of maximal density. In the case of simultaneous visibility in $\mathbb{Z}^k$, we realize the set $V(S)$ as a weak model set arising from a suitable cut-and-project set whose internal space is a countable product of finite sets over the prime numbers. Rumsey's density formula naturally appears as the Haar measure of the corresponding window. We then apply a theorem of Baake, Huck, and Strungaru [BHS17, Theorem 7] on the diffraction of weak model sets of maximal density to deduce that the diffraction spectrum is translation bounded pure point. The coefficients of the diffraction spectrum are computed by explicitly describing the dual lattice and the characters contained in it. We recall the result of Baake et al. in Theorem 2.1.

For visible points in cut-and-project sets associated with quadratic number fields, the strategy is similar but requires a more delicate construction of the internal space and window. Using results from [GKW25, Lemma 5.6], we characterize visibility in terms of arithmetic conditions and exclusion from a contracted copy of the window. This allows us to construct a new cut-and-project scheme whose associated weak model set of maximal density coincides with the set of visible points.

In both constructions, the corresponding windows are compact and have empty interior.

1.3. Remarks on the results and work in progress. Let K be a number field of degree $n = [K : \mathbb{Q}]$ whose ring of integers $\mathcal{O}_K$ is a principal ideal domain. Two distinct points $\mathbf{x} = (x_1, \dots, x_k)$ and $\mathbf{y} = (y_1, \dots, y_k)$ in $\mathcal{O}_K^k$ are said to be mutually visible if

$$\gcd(\mathbf{x} - \mathbf{y}) = \gcd((x_1 - y_1), \dots, (x_k - y_k)) = \mathcal{O}_K,$$

where $(x_i - y_i)$ denotes the ideal generated by $x_i - y_i$. For a finite set $S \subseteq \mathcal{O}_K^k$, denote by $V(S)$ the set of points that are simultaneously visible from every point in S . Analogous

to the $\mathbb{Z}^k$ -setting, this is defined by

$$V(S) = \{\mathbf{x} \in \mathcal{O}_K^k : \gcd(\mathbf{x} - \mathbf{y}) = \mathcal{O}_K \text{ for all } \mathbf{y} \in S\}.$$

The first-named author, in joint work [KT26], shows that the density of the set $V(S) \subset \mathcal{O}_K^k$ exists along some natural van Hove sequence. Using a similar argument for the proof of item 1 in Theorem 1.1, it is straightforward to verify that $V(S)$ is a weak model set of maximal density. Consequently, by Theorem 2.1, $\omega_{V(S)}$ is a translation-bounded pure point measure along natural van Hove sequences, provided the density of $V(S)$ is positive.

Furthermore, during the preparation of this manuscript, the authors learned of recent work by H. Shamir that generalizes Theorem 1.2. In particular, Shamir avoids the Hammarhjelm condition and instead only requires the real quadratic field $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$ with $\mathfrak{d} > 1$ square-free.

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2. PRELIMINARIES

We recall the main definitions and results needed for the proofs in this section, including the Hammarhjelm condition, van Hove sequences, diffraction spectra, cut-and-project sets and weak model sets.

2.1. The Hammarhjelm condition. Let $K = \mathbb{Q}(\sqrt{\mathfrak{d}})$ be a real quadratic field with square-free $\mathfrak{d} \geq 2$. Let $\mathcal{O}_K$ be its ring of integers. We assume throughout that $\mathcal{O}_K$ is a principal ideal domain. It is conjectured that there are infinitely many real quadratic number fields K for which $\mathcal{O}_K$ is a principal ideal domain (see [Neu99, p. 37]). The ring of integers $\mathcal{O}_K$ is a principal ideal domain for $2 \leq \mathfrak{d} \leq 100$

$$\begin{aligned} \mathfrak{d} = 2, 3, 5, 6, 7, 11, 13, 14, 17, 19, 21, 22, 23, 29, 31, 33, 37, 38, 41, 43, \\ 46, 47, 53, 57, 59, 61, 62, 67, 69, 71, 73, 77, 83, 86, 89, 93, 94, 97. \end{aligned}$$

For an ideal $I \subseteq \mathcal{O}_K$, the *norm* of I is defined by

$$N(I) \stackrel{\text{def}}{=} \#(\mathcal{O}_K/I),$$

which is always finite. We denote the Dedekind zeta function by

$$\zeta_{\mathcal{O}_K}(s) \stackrel{\text{def}}{=} \sum_{I \subseteq \mathcal{O}_K} \frac{1}{N(I)^s}, \quad \Re(s) > 1.$$

Denote by

$$\mathcal{P} = \{\mathfrak{p} \subset \mathcal{O}_K : \mathfrak{p} \text{ is prime and } 1 < \mathfrak{p} < \lambda\},$$

where λ is the fundamental unit.

A number field K *satisfies the Hammarhjelm condition* if $\mathcal{O}_K$ is a principal ideal domain and

$$|\sigma(\mathfrak{p})| > 1 \quad \text{for all } \mathfrak{p} \in \mathcal{P},$$

where σ is the non-trivial Galois automorphism of K and $|\cdot|$ denotes the absolute value. In the range $1 \leq \mathfrak{d} \leq 100$, the Hammarhjelm condition is satisfied only for $\mathfrak{d} = 2, 5, 13, 29, 53$; see [GKW25].

2.2. Van Hove sequences. Let G be a σ -compact, locally compact Abelian group (LCAG) with Haar measure θ_G . A sequence $\mathcal{A} = \{A_n\}_{n \in I}$ (typically $I = \mathbb{N}$ or $I = (0, \infty)$) of compact subsets of G is an *averaging sequence* if it is nested ($A_n \subseteq A_{n'}$ whenever $n < n'$) and satisfies $\bigcup_{n \in I} A_n = G$. Such a sequence is called a *van Hove sequence* if $\theta_G(A_n) > 0$ for all n and for every compact set $K \subseteq G$, the K -boundary is asymptotically negligible, that is,

$$\lim_{n \rightarrow \infty} \frac{\theta_G(\partial^K A_n)}{\theta_G(A_n)} = 0.$$

The set $\partial^K B$ denotes the K -boundary of B , defined by

$$\partial^K B = ((K + B) \setminus B^\circ) \cup ((-K + \overline{B^c}) \cap B),$$

where B^c , B° and $\overline{B}$ denote the complement, interior and closure of $B \subset G$, respectively (see [Moo02, p. 125], [Sch00, p. 145]).

For example, when $G = \mathbb{R}^d$, the sequence of boxes $A_n = [-n, n]^d$ and the sequence of balls centered at the origin with radius n , $A_n = B_n(0)$, are van Hove sequences. The existence of van Hove sequences in σ -compact LCAGs is shown in [Sch00].

The density of a point set $\Lambda \subset G$, calculated along an averaging sequence $\mathcal{A} = (A_n)_{n \in \mathbb{N}}$, is defined by

$$(5) \quad \theta(\Lambda) = \lim_{n \rightarrow \infty} \frac{\#(\Lambda \cap A_n)}{\theta_G(A_n)},$$

provided that the limit exists. The lower and upper densities are defined by

$$(6) \quad \underline{\theta}(\Lambda) = \liminf_{n \rightarrow \infty} \frac{\#(\Lambda \cap A_n)}{\theta_G(A_n)} \quad \text{and} \quad \bar{\theta}(\Lambda) = \limsup_{n \rightarrow \infty} \frac{\#(\Lambda \cap A_n)}{\theta_G(A_n)},$$

which always exist, with

$$0 \leq \underline{\theta}(\Lambda) \leq \bar{\theta}(\Lambda) \leq \infty.$$

2.3. Cut-and-project sets. Let G and H be σ -compact LCAGs. Denote their Haar measures by θ_G and θ_H , respectively. We assume that H is compactly generated. Hence, H is (up to isomorphism) of the form $\mathbb{R}^d \times \mathbb{Z}^n \times \mathbb{K}$, for some integers $d, n \geq 0$ and some compact Abelian group $\mathbb{K}$, see [HR79, Theorem 9.8].

A cut-and-project scheme $(G, H, \mathcal{L})$ consists of the product group $G \times H$, a full-rank lattice $\mathcal{L} \subset G \times H$ (that is, a discrete, cocompact subgroup), and the natural projections

$$\pi_{\text{phys}} : G \times H \longrightarrow G, \quad \pi_{\text{int}} : G \times H \longrightarrow H,$$

such that

- (I) The restriction $\pi_{\text{phys}}|_{\mathcal{L}} : \mathcal{L} \rightarrow G$ is injective;
- (D) The image $\pi_{\text{int}}(\mathcal{L})$ is dense in H .

Condition (I) allows us to define the $\star$ -map

$$(7) \quad (\cdot)^\star : \pi_{\text{phys}}(\mathcal{L}) \longrightarrow H, \quad x \longmapsto x^\star \stackrel{\text{def}}{=} \pi_{\text{int}}((\pi_{\text{phys}}|_{\mathcal{L}})^{-1}(x)).$$

Let $\mathcal{W} \subset H$ be a non-empty subset (typically assumed to be measurable), called the *window*. A cut-and-project set for the scheme $(G, H, \mathcal{L})$ is defined by

$$\Lambda(\mathcal{W}, \mathcal{L}) \stackrel{\text{def}}{=} \{\pi_{\text{phys}}(y) : y \in \mathcal{L}, \pi_{\text{int}}(y) \in \mathcal{W}\}.$$

Following [BHS17], a cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$ is called a *model set* if the window $\mathcal{W}$ is relatively compact and has non-empty interior, that is,

$$\mathcal{W}^\circ \neq \emptyset.$$

If $\mathcal{W} = \overline{\mathcal{W}^\circ}$ is compact, then the window is called *proper*. If $\mathcal{W}$ is relatively compact, has non-empty interior and satisfies

$$\theta_H(\partial\mathcal{W}) = 0,$$

then the cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$ is a model set, called *regular*. See [RSW24, § 2.1] for further discussion of this terminology. We use the terms *regular* and *irreducible* interchangeably, as they refer to the same condition in the literature; see [RSW24, § 2.1] and [BHS17, Page 6 and Remark 4]. Regular (equivalently, irreducible) cut-and-project sets are Delone sets (see [HR15, p. 744]).

Recall that a set $\mathcal{D} \subset G$ is *uniformly discrete* if there exists an open neighborhood $U \subset G$ of the identity such that any of its translates xU , where $x \in G$, contains at most one point of $\mathcal{D}$. A set $\mathcal{D} \subset G$ is *relatively dense* if there exists a compact set $K \subset G$ such that $\mathcal{D}K = G$. A *Delone set* $\mathcal{D} \subset G$ is a uniformly discrete and relatively dense set.

The density of an irreducible cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$ along a van Hove sequence $\mathcal{A} = (A_n)_{n \in \mathbb{N}}$ is given by

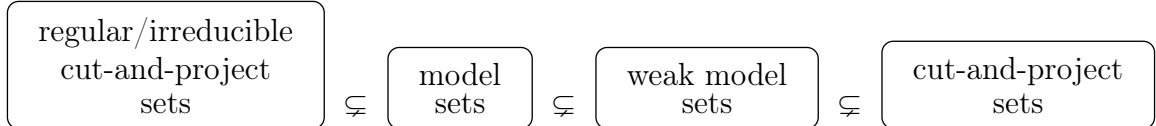
$$(8) \quad \theta(\Lambda(\mathcal{W}, \mathcal{L})) = \lim_{n \rightarrow \infty} \frac{\#(\Lambda(\mathcal{W}, \mathcal{L}) \cap A_n)}{\theta_G(A_n)} = \theta(\mathcal{L}) \cdot \theta_H(\mathcal{W}),$$

see [Moo02, Theorem 1], [Sch98] and [HR15, Lemma 3.2, Remark 3.3]. We refer to [Mey72, Chapter II] for further details on cut-and-project sets.

A cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$ is called a *weak model set* if the window $\mathcal{W}$ is relatively compact and satisfies

$$\theta_H(\overline{\mathcal{W}}) > 0.$$

Every weak model set is uniformly discrete (see [HR15, Proposition 2.4]). Moreover, if $\mathcal{W}$ is nowhere dense, then the cut-and-project set $\Lambda(\mathcal{W}, \mathcal{L})$ has arbitrarily large holes (see [HR15, Proposition 2.12]) and therefore is not relatively dense. We have the following inclusions:



2.4. Diffraction spectrum. Let G be a σ -compact LCAG. A measure ω on G is said to be *translation bounded* if, for every compact set $K \subseteq G$,

$$\sup_{t \in G} |\omega|(t + K) < \infty,$$

where $|\omega|$ denotes the total variation measure of ω . Throughout, measures are viewed as linear functionals on the space $C_c(G)$ of continuous functions with compact support. Given a measure μ on G and a function $g \in C_c(G)$, we write

$$\mu(g) \stackrel{\text{def}}{=} \int_G g \, d\mu.$$

Define the twist of a function g by

$$\tilde{g}(x) \stackrel{\text{def}}{=} \overline{g(-x)}$$

and the twist of a measure μ by

$$(9) \quad \tilde{\mu}(g) \stackrel{\text{def}}{=} \overline{\mu(\tilde{g})}, \quad g \in C_c(G).$$

Let μ and ν be measures on G . If μ is translation bounded and ν is finite, then the convolution $\mu * \nu$ is a measure defined by

$$(\mu * \nu)(f) \stackrel{\text{def}}{=} \iint_{G \times G} f(x + y) d\mu(x) d\nu(y), \quad f \in C_c(G).$$

See [MS17, Lemma 4.9.19] and [AGdL74, Theorem 1.2]. Recall that the dual group $\widehat{G}$ is the group of homomorphisms (characters)

$$\chi : G \longrightarrow \mathbb{S}^1,$$

equipped with pointwise multiplication and the compact-open topology. The group $\widehat{G}$ is again a LCAG. Let $\mathcal{A} = (A_n)_{n \in \mathbb{N}}$ be a van Hove sequence in G . The *autocorrelation* of ω along the sequence $\mathcal{A}$ is defined, provided the limit exists, as the weak-* limit

$$\gamma_\omega \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} \frac{\omega|_{A_n} * \widetilde{\omega|_{A_n}}}{\theta_G(A_n)}.$$

The Fourier transform of the autocorrelation, denoted by $\widehat{\gamma}_\omega$, is a positive, translation bounded measure on $\widehat{G}$. It is called the *diffraction spectrum* (or *diffraction measure*) of ω along the van Hove sequence $\mathcal{A}$. See [MS17, Theorems 4.10.14 and 4.11.7, and Lemma 4.11.3] for further details.

2.4.1. Further results on diffraction spectrum. We identify a character $\chi(\cdot) = \chi_u(\cdot)$ with its associated pairing $\langle u, \cdot \rangle$, so that in additive notation we have $\chi_u \chi_{u'} = \chi_{u+u'}$. There is a natural isomorphism $\widehat{G} \times \widehat{H} \cong \widehat{G} \times \widehat{H}$. To simplify the notation, denote the canonical projections on the dual groups by

$$\pi_{\text{phys}} : \widehat{G} \times \widehat{H} \longrightarrow \widehat{G}, \quad \pi_{\text{int}} : \widehat{G} \times \widehat{H} \longrightarrow \widehat{H}.$$

Given the lattice $\mathcal{L} \subset G \times H$, via the $\star$ -map defined in (7), we may write

$$\mathcal{L} = \{(x, x^\star) : x \in \pi_{\text{phys}}(\mathcal{L})\}.$$

Define $\mathcal{L}^0 \subset \widehat{G} \times \widehat{H}$ by

$$\mathcal{L}^0 \stackrel{\text{def}}{=} \left\{ (\chi_u, \chi_v) \in \widehat{G} \times \widehat{H} : \chi_u(x) \chi_v(x^\star) = 1 \text{ for all } (x, x^\star) \in \mathcal{L} \right\}.$$

The set $\mathcal{L}^0$ coincides with the annihilator of $\mathcal{L}$ in $\widehat{G \times H}$, which in turn implies that $\mathcal{L}^0$ is a lattice. If $(G, H, \mathcal{L})$ is a cut-and-project scheme, then the restriction $\pi_{\text{phys}}|_{\mathcal{L}^0}$ is injective, and the image $\pi_{\text{int}}(\mathcal{L}^0)$ is dense in $\widehat{H}$. This allows us to define the cut-and-project scheme $(\widehat{G}, \widehat{H}, \mathcal{L}^0)$. Following (7) and by abuse of notation, we define the $\star$ -map

$$(\cdot)^\star : \pi_{\text{phys}}(\mathcal{L}^0) \longrightarrow \widehat{H}, \quad \chi_u \longmapsto \chi_u^\star \stackrel{\text{def}}{=} \chi_v,$$

where $\chi_u : \widehat{G} \rightarrow \mathbb{S}^1$ and $\chi_v : \widehat{H} \rightarrow \mathbb{S}^1$ are as in the definition of $\mathcal{L}^0$, that is,

$$\chi_u(x) \chi_v(x^\star) = 1 \quad \text{for all } (x, x^\star) \in \mathcal{L}.$$

The *covariogram function* $c_{\mathcal{W}}$ of a relatively compact Borel set $\mathcal{W} \subset H$ is the real-valued function defined by

$$c_{\mathcal{W}}(x) = I_{\mathcal{W}} * \widetilde{I_{\mathcal{W}}}(x), \quad x \in H,$$

where $I_{\mathcal{W}}$ denotes the indicator function of $\mathcal{W}$. The convolution of two integrable functions $f, g \in L^1(H)$ is defined by

$$(f * g)(x) \stackrel{\text{def}}{=} \int_H f(y) g(x - y) d\theta_H(y).$$

It is a standard fact that $c_{\mathcal{W}}$ is bounded and uniformly continuous, since both $I_{\mathcal{W}}$ and $\widetilde{I}_{\mathcal{W}}$ are bounded integrable functions on H .

2.5. Weak model sets of maximal density. Let $\Lambda(\mathcal{W}, \mathcal{L})$ be a cut-and-project set, where G and H are σ -compact LCAGs and H is compactly generated. We say that $\Lambda(\mathcal{W}, \mathcal{L})$ is a *weak model set of maximal density* relative to a given van Hove averaging sequence $\mathcal{A}$ if $\Lambda(\mathcal{W}, \mathcal{L})$ is a weak model set (recall that this means $\mathcal{W} \subset H$ is relatively compact and $\theta_H(\overline{\mathcal{W}}) > 0$), and if the density of $\Lambda(\mathcal{W}, \mathcal{L})$ relative to $\mathcal{A}$ exists and is given by

$$\theta(\Lambda(\mathcal{W}, \mathcal{L})) = \theta(\mathcal{L}) \cdot \theta_H(\overline{\mathcal{W}}).$$

We will use the following result.

Theorem 2.1. [BHS17, Theorem 7] *Let $\Lambda = \Lambda(\mathcal{W}, \mathcal{L})$ be a cut-and-project set with compact window $\mathcal{W} \subset H$ and $\theta_H(\mathcal{W}) > 0$. If Λ is a weak model set of maximal density, then the diffraction spectrum $\widehat{\gamma}_{\Lambda}$ is translation bounded pure point. Moreover, it is given by*

$$\widehat{\gamma}_{\Lambda} = \sum_{\chi \mathbf{u} \in \pi_{\text{phys}}(\mathcal{L}^0)} |a(\chi \mathbf{u})|^2 \delta_{\chi \mathbf{u}},$$

where

$$a(\chi \mathbf{u}) = \frac{\theta(\Lambda)}{\theta_H(\mathcal{W})} \widehat{I_{\mathcal{W}}(-\chi \mathbf{u}^*)},$$

and $\widehat{I_{\mathcal{W}}}$ is a bounded, continuous function on $\widehat{H}$.

3. PROOFS

3.1. Proof of Theorem 1.1 Item 1. Let $S \subseteq \mathbb{Z}^k$ be a finite set, with $k \geq 2$. We show that $V(S)$, the set of points in $\mathbb{Z}^k$ simultaneously visible from every point of S , is a weak model set of maximal density. Then Theorem 2.1 implies that the diffraction spectrum $\widehat{\gamma_{\omega_{V(S)}}$ is a translation bounded pure point measure.

We work with

$$G = \mathbb{Z}^k, \quad H = \prod_{p \in \mathcal{P}} (\mathbb{Z}/p\mathbb{Z})^k,$$

where H is equipped with the product topology and the product normalized Haar probability measure θ_H , so that each factor $(\mathbb{Z}/p\mathbb{Z})^k$ carries the normalized counting measure of total mass 1. The groups G and H are σ -compact LCAGs, and H is compactly generated. Fix the van Hove sequence $\mathcal{A} = \{[-L, L]^k\}_{L \in \mathbb{N}}$.

Define the set $\mathcal{L}$ by

$$\mathcal{L} \stackrel{\text{def}}{=} \{(\mathbf{x}, \iota(\mathbf{x})) : \mathbf{x} \in \mathbb{Z}^k\} \subset G \times H,$$

where

$$\iota(\mathbf{x}) \stackrel{\text{def}}{=} (\pi_p(\mathbf{x}))_{p \in \mathcal{P}}$$

and $\pi_p : \mathbb{Z}^k \rightarrow (\mathbb{Z}/p\mathbb{Z})^k$ is the natural projection defined in (1).

The set $\mathcal{L}$ is a discrete subgroup of $G \times H$. Moreover, $\iota(\mathbb{Z}^k)$ is dense in H by the Chinese Remainder Theorem. Since the map

$$H \longrightarrow (G \times H)/\mathcal{L}, \quad h \longmapsto (0, h) + \mathcal{L}$$

is continuous and surjective and H is compact, it follows that $(G \times H)/\mathcal{L}$ is compact. Hence, $\mathcal{L}$ is a cocompact lattice in $G \times H$.

Let $\pi_{\text{phys}} : G \times H \rightarrow G$ and $\pi_{\text{int}} : G \times H \rightarrow H$ be the projections. Note that $\pi_{\text{phys}}|_{\mathcal{L}}$ is injective and $\pi_{\text{int}}(\mathcal{L}) = \iota(\mathbb{Z}^k)$ is dense in H . Hence $(G, H, \mathcal{L})$ is a cut-and-project scheme.

For each $p \in \mathcal{P}$, define

$$\mathcal{W}_p \stackrel{\text{def}}{=} (\mathbb{Z}/p\mathbb{Z})^k \setminus \pi_p(S)$$

and define

$$\mathcal{W}_S \stackrel{\text{def}}{=} \prod_{p \in \mathcal{P}} \mathcal{W}_p \subset H.$$

Each $\mathcal{W}_p$ is finite and therefore compact. Hence, by Tychonoff's theorem, $\mathcal{W}_S$ is a compact subset of H . By construction, a vector $\mathbf{x} \in \mathbb{Z}^k$ satisfies $\iota(\mathbf{x}) \in \mathcal{W}_S$ if and only if for every prime p , the residue class of $\mathbf{x}$ modulo p does not belong to $\pi_p(S)$. This is equivalent to $\mathbf{x}$ being simultaneously visible from all points of S . Therefore, the cut-and-project set

$$\Lambda(\mathcal{W}_S, \mathcal{L}) \stackrel{\text{def}}{=} \{\pi_{\text{phys}}(\mathbf{y}) : \mathbf{y} \in \mathcal{L}, \pi_{\text{int}}(\mathbf{y}) \in \mathcal{W}_S\} = \{\mathbf{x} \in \mathbb{Z}^k : \iota(\mathbf{x}) \in \mathcal{W}_S\}$$

coincides with $V(S)$. The normalized measure of $\mathcal{W}_p$ is

$$\theta_{H,p}(\mathcal{W}_p) = 1 - \frac{s(p)}{p^k},$$

where $\theta_{H,p}$ is the normalized Haar measure on $(\mathbb{Z}/p\mathbb{Z})^k$. Hence, by the product measure,

$$(10) \quad \theta_H(\mathcal{W}_S) = \prod_{p \in \mathcal{P}} \left(1 - \frac{s(p)}{p^k}\right).$$

This product coincides with Rumsey's density formula (2) of the point set $V(S)$. By the assumption that the density of $V(S)$ is positive, we get $\theta_H(\overline{\mathcal{W}_S}) = \theta_H(\mathcal{W}_S) > 0$ and $\Lambda(\mathcal{W}_S, \mathcal{L}) = V(S)$. By (2) and (10), we get

$$\theta(\Lambda(\mathcal{W}_S, \mathcal{L})) = \theta(V(S)) = \theta_H(\mathcal{W}_S) = \theta(\mathcal{L})\theta_H(\overline{\mathcal{W}_S}),$$

since $\theta(\mathcal{L}) = 1$. Therefore, $V(S)$ is a weak model set of maximal density. Hence, Theorem 2.1 implies that $\widehat{\gamma_{\omega_{V(S)}}}$ is a translation bounded pure point measure.

3.2. Proof of Theorem 1.1 Item 2. Recall that

$$G = \mathbb{Z}^k, \quad H = \prod_{p \in \mathcal{P}} H_p, \quad H_p = (\mathbb{Z}/p\mathbb{Z})^k,$$

$$\mathcal{W}_S = \prod_{p \in \mathcal{P}} \mathcal{W}_p, \quad \mathcal{W}_p = H_p \setminus \pi_p(S),$$

and the lattice $\mathcal{L}$ is given by

$$\mathcal{L} \stackrel{\text{def}}{=} \{(\mathbf{x}, \mathbf{x}^*) : \mathbf{x} \in \mathbb{Z}^k\} \subset G \times H,$$

where $\mathbf{x}^* \stackrel{\text{def}}{=} (\pi_p(\mathbf{x}))_{p \in \mathcal{P}}$. We use the notation $\mathbf{x}_p$ and $\pi_p(\mathbf{x})$ interchangeably to shorten the expressions. The dual lattice of $\mathcal{L}$ has been calculated (see for example [BLS26, p.6] and [RS17, p.14]). For self-containment, we will recall the construction. We begin by setting up the notation for the dual groups. Recall that

$$\widehat{\mathbb{Z}} = \left\{m : x \mapsto \chi_m(x) \stackrel{\text{def}}{=} \exp(2\pi i x m), \forall x \in \mathbb{Z}\right\} = \mathbb{R}$$

where $\exp(x) \stackrel{\text{def}}{=} e^x$ is the exponential function. Since $\widehat{G} = \widehat{\mathbb{Z}^k} \cong (\widehat{\mathbb{Z}})^k$, it follows that

$$\widehat{G} \cong \left\{\mathbf{m} : \mathbf{x} \mapsto \chi_{\mathbf{m}}(\mathbf{x}) \stackrel{\text{def}}{=} \exp(2\pi i \langle \mathbf{x}, \mathbf{m} \rangle) \quad \forall \mathbf{x} \in \mathbb{Z}^k\right\} = \mathbb{R}^k.$$

For each prime $p \in \mathcal{P}$, the dual group of $\mathbb{Z}/p\mathbb{Z}$ is

$$\widehat{\mathbb{Z}/p\mathbb{Z}} = \left\{n : y \mapsto \chi_n(y) \stackrel{\text{def}}{=} \exp\left(2\pi i \frac{yn}{p}\right), \forall y \in \mathbb{Z}/p\mathbb{Z}\right\} = \{0, 1, \dots, p-1\} = \mathbb{Z}/p\mathbb{Z}.$$

Hence

$$(\widehat{\mathbb{Z}/p\mathbb{Z}})^k \cong \left\{\mathbf{n} : \mathbf{y} \mapsto \chi_{\mathbf{n}}(\mathbf{y}) \stackrel{\text{def}}{=} \exp\left(2\pi i \frac{\langle \mathbf{y}, \mathbf{n} \rangle}{p}\right)\right\} = \{0, 1, \dots, p-1\}^k = (\mathbb{Z}/p\mathbb{Z})^k.$$

The dual group $\hat{H}$ is isomorphic to the direct sum

$$\hat{H} \cong \bigoplus_{p \in \mathcal{P}} (\widehat{\mathbb{Z}/p\mathbb{Z}})^k.$$

An element $(\mathbf{n}_p)_{p \in \mathcal{P}} \in \hat{H}$ is a sequence of *finitely many* non-zero vectors. Such sequence represents a character $\chi_{(\mathbf{n}_p)_{p \in \mathcal{P}}}$ defined by

$$(\mathbf{y}_p)_{p \in \mathcal{P}} \mapsto \chi_{(\mathbf{n}_p)_{p \in \mathcal{P}}}((\mathbf{y}_p)_{p \in \mathcal{P}}) \stackrel{\text{def}}{=} \prod_{p \in \mathcal{P}} \exp\left(2\pi i \frac{\langle \mathbf{y}_p, \mathbf{n}_p \rangle}{p}\right), \quad \text{for all } (\mathbf{y}_p)_{p \in \mathcal{P}} \in H.$$

Note that the character is well defined, since only finitely many $\mathbf{n}_p$'s are non-zero. Hence the dual lattice of $\mathcal{L}$ consists of points $(\mathbf{u}, (\mathbf{n}_p)_{p \in \mathcal{P}}) \in \hat{G} \times \hat{H}$ such that

$$(11) \quad \exp(2\pi i \langle \mathbf{x}, \mathbf{u} \rangle) \cdot \prod_{p \in \mathcal{P}} \exp\left(2\pi i \frac{\langle \mathbf{x}_p, \mathbf{n}_p \rangle}{p}\right) = 1, \quad \text{for all } (\mathbf{x}, \mathbf{x}^*) \in \mathcal{L}.$$

Because of linearity and the rules of modular arithmetic, we only need to verify this condition for the canonical basis vectors $\mathbf{x} \in \{e_1, \dots, e_k\}$ of $\mathbb{Z}^k$. By writing the vectors in their component forms as $\mathbf{u} = (u_1, \dots, u_k)$ and $\mathbf{n}_p = (n_{1,p}, \dots, n_{k,p})$, eq. (11) breaks down into the following coordinate-by-coordinate requirements

$$(12) \quad u_j = \frac{m_j}{N_j} - \sum_{p \in \mathcal{P}} \frac{n_{j,p}}{p}, \quad \text{for all } j \in 1, \dots, k,$$

where m_j, N_j are integers and $n_{j,p} \in \{0, \dots, p-1\}$. Consequently, if $(\mathbf{u}, (\mathbf{n}_p)_{p \in \mathcal{P}}) \in \mathcal{L}^0$, then $(\mathbf{u} + \mathbf{t}, (\mathbf{n}_p)_{p \in \mathcal{P}}) \in \mathcal{L}^0$ for every translation vector $\mathbf{t} \in \mathbb{Z}^k$. Given that only a finite number of $\mathbf{n}_p$'s are non-zero, the individual conditions of eq. (12) can be algebraically manipulated to share a least common denominator (lcm). Combining all these conditions reveals that they are equivalent to

$$\pi_{\text{phys}}(\mathcal{L}^0) = \left\{ \mathbf{u} = \frac{\mathbf{m}}{N} \in \mathbb{Q}^k : N \text{ is square-free, } \mathbf{m} \in \mathbb{Z}^k \right\}.$$

Moreover, the $\star$ -map for the dual lattice is given by

$$\mathbf{u} = \frac{\mathbf{m}}{N} \mapsto \mathbf{u}^\star := \left(-\text{lcm}(N, p) \cdot \frac{\mathbf{m}}{N} \pmod{p} \right)_{p \in \mathcal{P}}.$$

In which case, the entry at p is $-\mathbf{m} \pmod{p}$ for all $p \mid N$ and otherwise 0. Let $\mathbf{v}_p$ denote the expression $-\text{lcm}(N, p) \frac{\mathbf{m}}{N} \pmod{p}$ for each $p \in \mathcal{P}$.

By Theorem 2.1, and using the fact that $\theta(V(S)) = \theta_H(\mathcal{W}_S)$, we have that for all $\mathbf{u} \in \pi_{\text{phys}}(\mathcal{L}^0)$,

$$a(\mathbf{u}) = \frac{\theta(V(S))}{\theta_H(\mathcal{W}_S)} \cdot \widehat{I_{\mathcal{W}_S}}(-\mathbf{u}^\star) = \widehat{I_{\mathcal{W}_S}}(-\mathbf{u}^\star).$$

By the previous discussion about $\hat{H}$, $\mathbf{u}^\star = (\mathbf{v}_p)_{p \in \mathcal{P}}$ acts via

$$\prod_{p \in F} \exp\left(2\pi i \frac{\langle \mathbf{v}_p, \cdot \rangle}{p}\right)$$

over a finite set $F \subset \mathcal{P}$. If $F = \emptyset$, the character is constant and equal to 1 and if $p \notin F$, then $\mathbf{v}_p = \mathbf{0}$.

Now, $I_{\mathcal{W}_S} = \prod_{p \in \mathcal{P}} I_{\mathcal{W}_p}$, so the Fourier coefficient factors prime by prime, that is

$$\widehat{I_{\mathcal{W}_S}}(-\mathbf{u}^\star) = \prod_{p \in \mathcal{P}} \widehat{I_{\mathcal{W}_p}}(-\mathbf{v}_p).$$

For each prime p ,

$$\begin{aligned}
 \widehat{I_{\mathcal{W}_p}}(-\mathbf{v}_p) &= \frac{1}{p^k} \sum_{\mathbf{x} \in H_p \setminus \pi_p(S)} \exp\left(\frac{-2\pi i \langle \mathbf{v}_p, \mathbf{x} \rangle}{p}\right) \\
 (13) \quad &= \frac{1}{p^k} \left(\sum_{\mathbf{x} \in H_p} \exp\left(\frac{-2\pi i \langle \mathbf{v}_p, \mathbf{x} \rangle}{p}\right) - \sum_{\mathbf{x} \in \pi_p(S)} \exp\left(\frac{-2\pi i \langle \mathbf{v}_p, \mathbf{x} \rangle}{p}\right) \right).
 \end{aligned}$$

Now compute the full character sum. If $\mathbf{v}_p = \mathbf{0}$, then every exponential is 1, so

$$\sum_{\mathbf{x} \in H_p} \exp\left(-2\pi i \frac{\langle \mathbf{0}, \mathbf{x} \rangle}{p}\right) = \#(H_p) = p^k,$$

and therefore

$$\widehat{I_{\mathcal{W}_p}}(\mathbf{0}) = \frac{p^k - s(p)}{p^k} = 1 - \frac{s(p)}{p^k}.$$

If $\mathbf{v}_p \neq \mathbf{0}$, choose j such that the j -th coordinate of $\mathbf{v}_p$ is non-zero, that is $\mathbf{v}_{p,j} \neq 0$. Since $\mathbf{x} \in \{0, \dots, p-1\}^k$, we get

$$\sum_{\mathbf{x} \in H_p} \exp\left(-2\pi i \frac{\langle \mathbf{v}_p, \mathbf{x} \rangle}{p}\right) = \prod_{r=1}^k \left(\sum_{n=0}^{p-1} \exp\left(-2\pi i \frac{\mathbf{v}_{p,r} n}{p}\right) \right).$$

In the j -th coordinate, the ratio $\exp\left(-2\pi i \frac{\mathbf{v}_{p,j} n}{p}\right) \neq 1$ is a non-trivial p -th root of unity, that is $\sum_{n=0}^{p-1} \exp\left(-2\pi i \frac{\mathbf{v}_{p,j} n}{p}\right) = 0$. Hence, the whole product is 0. Therefore

$$\widehat{I_{\mathcal{W}_p}}(-\mathbf{v}_p) = -\frac{1}{p^k} \sum_{\mathbf{x} \in \pi_p(S)} \exp\left(-2\pi i \frac{\langle \mathbf{v}_p, \mathbf{x} \rangle}{p}\right), \quad \mathbf{v}_p \neq \mathbf{0}.$$

So the local formula is

$$(14) \quad \widehat{I_{\mathcal{W}_p}}(-\mathbf{v}_p) = \begin{cases} 1 - \frac{s(p)}{p^k}, & \mathbf{v}_p = \mathbf{0}, \\ -\frac{1}{p^k} \sum_{\mathbf{x} \in \pi_p(S)} \exp\left(-2\pi i \frac{\langle \mathbf{v}_p, \mathbf{x} \rangle}{p}\right), & \mathbf{v}_p \neq \mathbf{0}. \end{cases}$$

Let $\mathbf{u} = \mathbf{m}/N \in \pi_{\text{phys}}(\mathcal{L}^0)$. It follows from the definition of $\mathcal{L}^0$ that only prime divisors of N can contribute non-trivially. Therefore,

$$\widehat{I_{\mathcal{W}_S}}(-u^\star) = \prod_{p \nmid N} \left(1 - \frac{s(p)}{p^k}\right) \cdot \prod_{p \mid N} \left(-\frac{1}{p^k} \sum_{\mathbf{x} \in \pi_p(S)} \exp\left(-2\pi i \frac{\langle \mathbf{v}_p, \mathbf{x} \rangle}{p}\right)\right).$$

Since

$$\theta_H(\mathcal{W}_S) = \prod_{p \in \mathcal{P}} \left(1 - \frac{s(p)}{p^k}\right) \quad \text{and} \quad \theta(V(S)) = \theta_H(\mathcal{W}_S),$$

we get

$$\begin{aligned}
 \widehat{I_{\mathcal{W}_S}}(-\mathbf{u}^\star) &= \theta_H(\mathcal{W}_S) \prod_{p|N} \frac{\left(-\frac{1}{p^k} \sum_{\mathbf{x} \in \pi_p(S)} \exp \left(-2\pi i \frac{\langle \mathbf{v}_p, \mathbf{x} \rangle}{p} \right) \right)}{\left(1 - \frac{s(p)}{p^k} \right)} \\
 (15) \qquad &= \theta(V(S)) \prod_{\substack{p \in \mathcal{P} \\ p|N}} \frac{\left(-\frac{1}{p^k} \sum_{\mathbf{x} \in \pi_p(S)} \exp \left(-2\pi i \frac{\langle \mathbf{v}_p, \mathbf{x} \rangle}{p} \right) \right)}{\left(1 - \frac{s(p)}{p^k} \right)}.
 \end{aligned}$$

3.3. Proof of Theorem 1.3. Let $\Lambda = \Lambda(\mathcal{W}, \mathcal{L})$ be a cut-and-project set as in the statement of Theorem 1.3. We show that $\Lambda_{\text{vis}} = \Lambda(\mathcal{W}, \mathcal{L})_{\text{vis}}$ is a weak model set of maximal density; then Theorem 2.1 implies translation bounded pure point diffraction.

Let σ be the non-trivial Galois automorphism of K . By abuse of notation $\sigma(x_1, \dots, x_d)$ means $(\sigma(x_1), \dots, \sigma(x_d))$. It follows from [GKW25, Lemma 5.6] that

$$(16) \quad \Lambda_{\text{vis}} = \left\{ \mathbf{x} = (x_1, \dots, x_d) \in \Lambda(\mathcal{W}, \mathcal{L})_\star : \gcd(x_1, \dots, x_d) = \mathcal{O}_K, \sigma(\mathbf{x}) \notin \frac{1}{\lambda} \mathcal{W} \right\}.$$

This representation allows us to construct a new cut-and-project scheme whose associated weak model set is Λ_{vis} . Define

$$\mathcal{P}_{\mathcal{O}_K} \stackrel{\text{def}}{=} \{ \mathfrak{p} \subset \mathcal{O}_K : \mathfrak{p} \text{ is prime ideal} \}.$$

Let $G = \mathbb{R}^d$ be the physical space and let

$$H \stackrel{\text{def}}{=} \mathbb{R}^d \times \prod_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} (\mathcal{O}_K/\mathfrak{p})^d$$

be the internal space with the corresponding projection maps π_{phys} and π_{int} . Both G and H are σ -compact LCAGs. The group H is compactly generated.

Let m be the Minkowski-type embedding

$$m : \mathcal{O}_K^d \hookrightarrow G \times H, \quad \mathbf{x} \mapsto (\mathbf{x}, \sigma(\mathbf{x}), (\pi_{\mathfrak{p}}(\mathbf{x}))_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}}),$$

where

$$\pi_{\mathfrak{p}} : \mathcal{O}_K^d \rightarrow (\mathcal{O}_K/\mathfrak{p}\mathcal{O}_K)^d, \quad \mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}$$

is the quotient map. Define the set $\mathcal{L}' \stackrel{\text{def}}{=} m(\mathcal{O}_K^d)$. The restriction $\pi_{\text{phys}}|_{\mathcal{L}'} : \mathcal{L}' \rightarrow G$ is injective since the map $\sigma(\cdot)$ determines $\mathbf{x}$ uniquely. Moreover, the image $\pi_{\text{int}}(\mathcal{L}')$ is dense in H , because the first factor of H is $\mathbb{R}^d$ and the second factor $\prod_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} (\mathcal{O}_K/\mathfrak{p})^d$ has the product topology. The quotient $G \times H/\mathcal{L}'$ is isomorphic to $\mathbb{R}^{2d}/\{(\mathbf{x}, \sigma(\mathbf{x})) : \mathbf{x} \in \mathcal{O}_K^d\}$. Thus, $\mathcal{L}' \subset G \times H$ is a cocompact lattice.

Define

$$\mathcal{W}'_\infty \stackrel{\text{def}}{=} \mathcal{W} \setminus \left(\frac{1}{\lambda} \mathcal{W} \right) = \left\{ w \in \mathcal{W} : w \notin \frac{1}{\lambda} \mathcal{W} \right\}$$

and for each prime ideal $\mathfrak{p} \subset \mathcal{O}_K$, define

$$\mathcal{W}'_{\mathfrak{p}} \stackrel{\text{def}}{=} (\mathcal{O}_K/\mathfrak{p}\mathcal{O}_K)^d \setminus \{\mathbf{0}\}.$$

Finally, define

$$\mathcal{W}' \stackrel{\text{def}}{=} \mathcal{W}'_\infty \times \prod_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} \mathcal{W}'_{\mathfrak{p}}.$$

which satisfies that $\mathcal{W}' = \overline{\mathcal{W}'}$. The cut-and-project set associated with this window is

$$\Lambda' = \Lambda'(\mathcal{W}', \mathcal{L}') = \{\pi_{\text{phys}}(l) : l \in \mathcal{L}', \pi_{\text{int}}(l) \in \mathcal{W}'\}.$$

The set Λ' coincides with Λ_{vis} . Indeed, the condition $\pi_{\text{int}}(l) \in \mathcal{W}'$ is equivalent to

- (1) $\mathbf{x} \stackrel{\text{def}}{=} \pi_{\text{phys}}(l) \in \mathcal{O}_K^d$ satisfying $\sigma(\mathbf{x}) \in \mathcal{W}'_{\infty}$, that is, $\mathbf{x} \in \Lambda$ and $\sigma(\mathbf{x}) \notin \frac{1}{\lambda}\mathcal{W}$ and
- (2) for every prime ideal $\mathfrak{p} \subset \mathcal{O}_K$, we have $\mathbf{x} \not\equiv 0 \pmod{\mathfrak{p}}$ in $(\mathcal{O}_K/\mathfrak{p}\mathcal{O}_K)^d$, which is equivalent to $\gcd(\mathbf{x}) = \mathcal{O}_K$.

Next, we compute the measure of $\mathcal{W}'$ in H . Each finite factor $\mathcal{W}'_{\mathfrak{p}}$, with $\mathfrak{p} \subset \mathcal{O}_K$ a prime ideal, has normalized measure

$$\theta_{H,\mathfrak{p}}(\mathcal{W}'_{\mathfrak{p}}) = 1 - \frac{1}{N(\mathfrak{p})^d},$$

where $\theta_{H,\mathfrak{p}}$ is the Haar probability measure on $(\mathcal{O}_K/\mathfrak{p}\mathcal{O}_K)^d$. Therefore,

$$\theta_H(\mathcal{W}') = \text{vol}(\mathcal{W}'_{\infty}) \cdot \prod_{\mathfrak{p} \in \mathcal{P}_{\mathcal{O}_K}} \left(1 - \frac{1}{N(\mathfrak{p})^d}\right) = \left(1 - \frac{1}{\lambda^d}\right) \text{vol}(\mathcal{W}) \cdot \frac{1}{\zeta_{\mathcal{O}_K}(d)}.$$

Since $d \geq 2$, we have $\theta_H(\overline{\mathcal{W}'}) > 0$. The cut-and-project set $\Lambda'(\mathcal{W}', \mathcal{L}')$ is a weak model set. Given a Jordan measurable set D , fix the van Hove sequence $\{TD\}_{T>0}$. It follows from Theorem 1.2 that $\Lambda'(\mathcal{W}', \mathcal{L}')$ is a weak model set of maximal density. By Theorem 2.1, the diffraction spectrum of Λ_{vis} is translation bounded pure point.

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